

$$\Delta H = ?$$

Calculate the change in Enthalpy for one mole of Hydrogen gas going from state-I (1×10^5 Pa, 300 K) to state-II (2×10^5 Pa, 600 K). Assuming "a" to be negligible and "b" as 50 ml mol^{-1} . C_p for the gas is $20 \text{ JK}^{-1} \text{ mol}^{-1}$.

[Given : $\left(\frac{\partial H}{\partial P}\right)_T = -T \left(\frac{\partial V}{\partial T}\right)_P + V$]

(A) 6000 J

(B) 11000 J

(C) 1000 J

(D) 6005 J

For a real gas, $H = f(T, P)$

$$dH = \left(\frac{\partial H}{\partial T}\right)_P dT + \left(\frac{\partial H}{\partial P}\right)_T dP$$

$$dH = C_p dT + \left(\frac{\partial H}{\partial P}\right)_T dP \quad \text{--- (1)}$$

$$\begin{aligned} dH &= C_p dT - \left(\frac{RT}{P}\right) dP + v dp \\ &= C_p dT - (v - b) dp + v dp \\ &= C_p dT + b dp \end{aligned}$$

$$\begin{aligned} dH &= C_p dT - T \left(\frac{\partial v}{\partial T}\right)_P dP + v dp \quad \text{--- (2)} \\ \left(P + \frac{a}{V^2}\right)(v - b) &= RT \end{aligned}$$

for 1 mol $P(v - b) = RT$

$$v = b + \frac{RT}{P}$$

$$\Delta H = C_p (T_2 - T_1) + b(P_2 - P_1) = 20 \times (600 - 300) + 50 \times 10^{-6} \left(\frac{\partial v}{\partial T} \right)_P = 0 + \frac{R}{P} \int_{1 \times 10^5}^{2 \times 10^5} = 6000 + 5 = 6005 \text{ Joule}$$

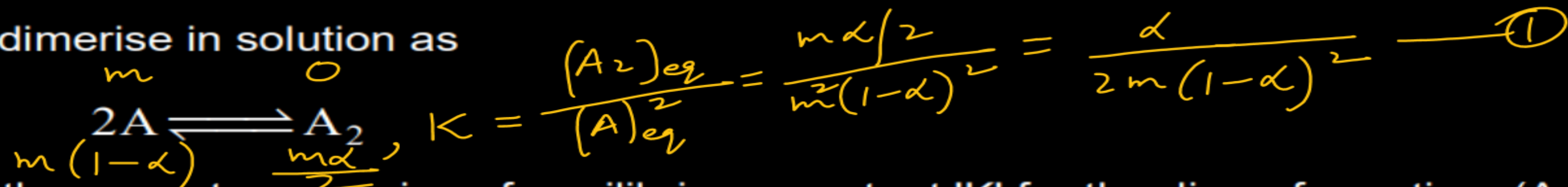
V. Imp

$H = f(T)$
 $U = f(T)$ } for an ideal gas.

$$\begin{aligned} \Delta H &= n \int_1^2 C_p dT \\ \Delta U &= n \int_1^2 C_v dT \end{aligned}$$

Ideal gases.

A dilute solution contains m mole of solute A in 1 kg of a solvent with molal elevation constant K_b . The solute dimerise in solution as



Select the correct expression of equilibrium constant 'K' for the dimer formation. (Assume molarity = molality)

(A) $K = \frac{K_b(K_b m - \Delta T_b)}{(K_b m + 2\Delta T_b)}$ ~~(B) $K = \frac{\alpha}{2m(1-\alpha)^2}$~~ ~~(C) $K = \frac{K_b(K_b m - \Delta T_b)}{(2\Delta T_b - K_b m)^2}$~~ (D) $K = \frac{2(K_b m - \Delta T_b)}{K_b m}$

$\Delta T_b = i \times K_b \times \text{molality}$

$\Delta T_b = (1 - \alpha/2) \times K_b \times m / 1 \text{ kg}$

$1 - \alpha/2 = \frac{\Delta T_b}{K_b m}$

$\Rightarrow \alpha/2 = \frac{(K_b m - \Delta T_b)}{K_b m} \Rightarrow \alpha = \frac{2(K_b m - \Delta T_b)}{K_b m}$

$i = 1 + \left(\frac{1}{n} - 1\right) \times \alpha, \quad n = 2$

$= (1 - \alpha/2)$

$K = \frac{\frac{2(K_b m - \Delta T_b)}{K_b m}}{\left(\frac{2(K_b m - \Delta T_b)}{K_b m}\right)^2} = \frac{K_b m}{2(K_b m - \Delta T_b)}$

$$a \text{ \AA} = a \times 10^{-8} \text{ cm}$$

The hexagonal close-packed lattice can be represented by the given figure, and $c = \left(\sqrt{\frac{8}{3}} \right) a$. There is

HCP

an atom at each corner of the unit cell and another atom which can be located by moving one-third the distance along the diagonal of the rhombus base, starting at the lower left-hand corner and moving perpendicularly upward by $c/2$. Cu crystallizes in this lattice and has a density of 1.8 g/cm^3 . Then the distance between nearest neighbours is: $\rightarrow = 2a = a$

distance between 1st and 3rd layer.

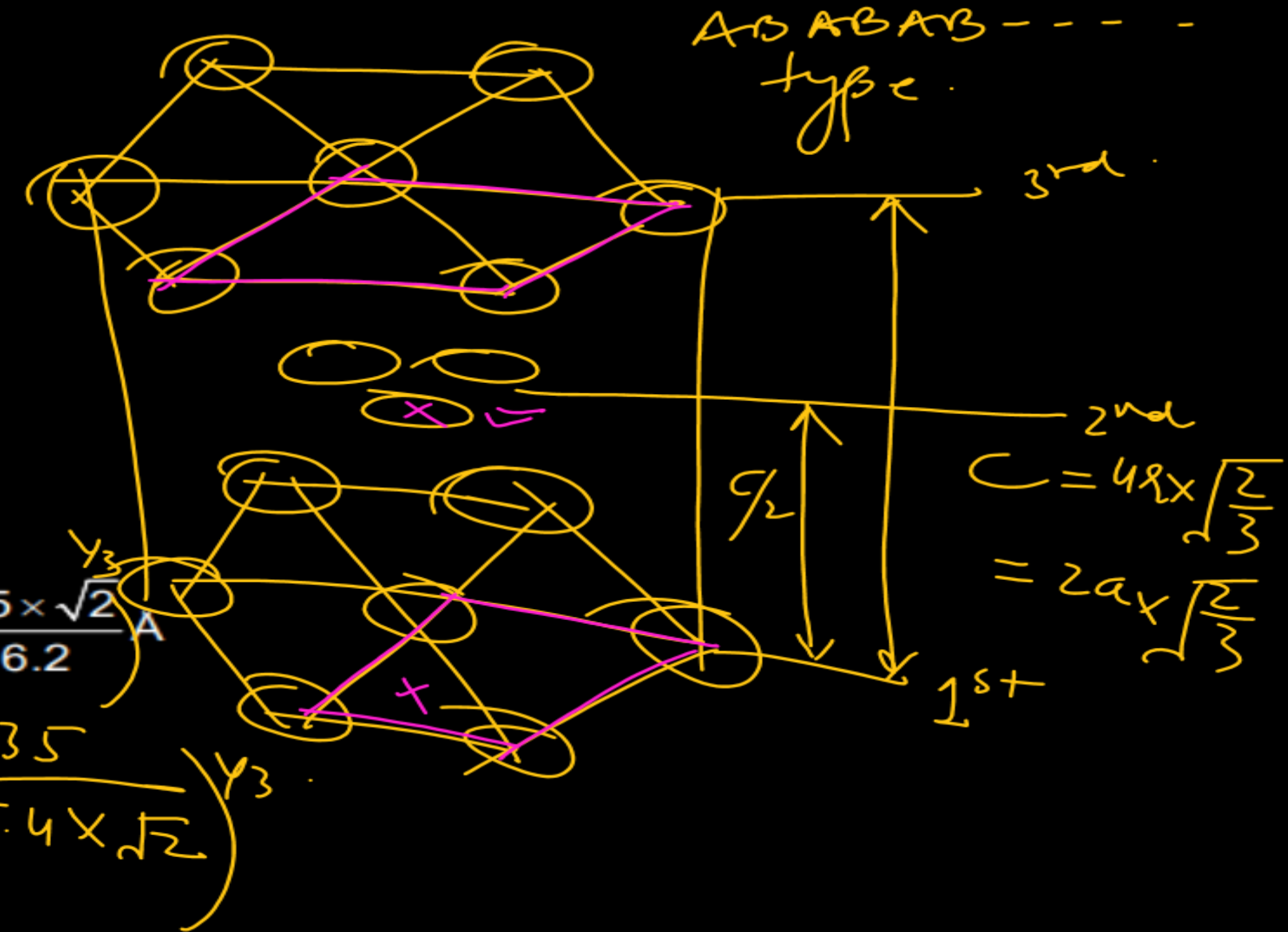
$d = 1.8 = \left(6 \times \frac{63.5}{6 \times 10^{23}} \right) \times \frac{6 \times \left(\frac{\sqrt{3}}{4} a^2 \right) \times c}{10^{23} \times \frac{\sqrt{3}}{4} \times a^2 \times \frac{2\sqrt{2}}{\sqrt{3}} \times a^3}$

$1.8 = \frac{63.5}{10^{23} \times \frac{\sqrt{3}}{4} \times a^2 \times \frac{2\sqrt{2}}{\sqrt{3}} \times a^3}$

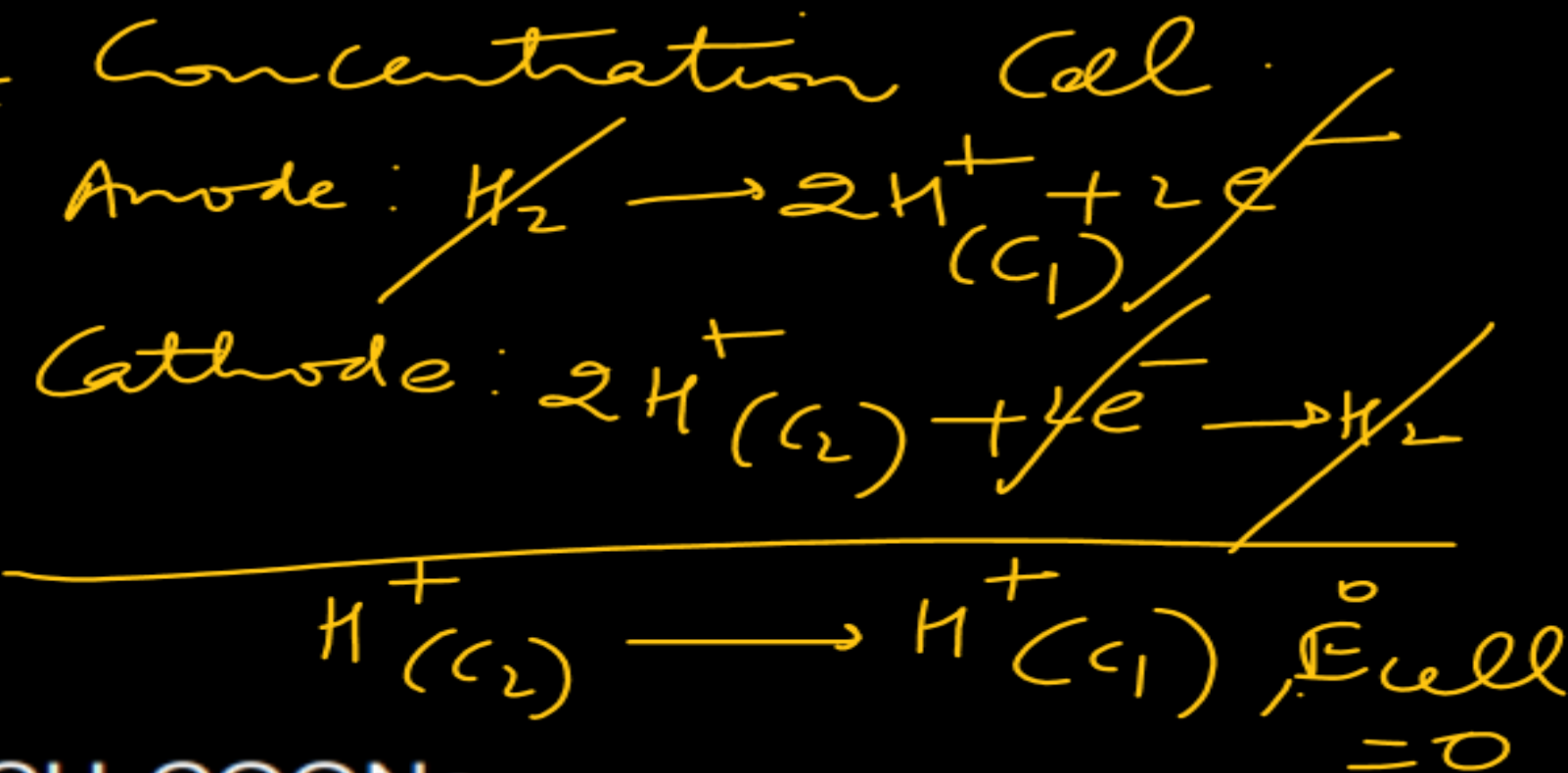
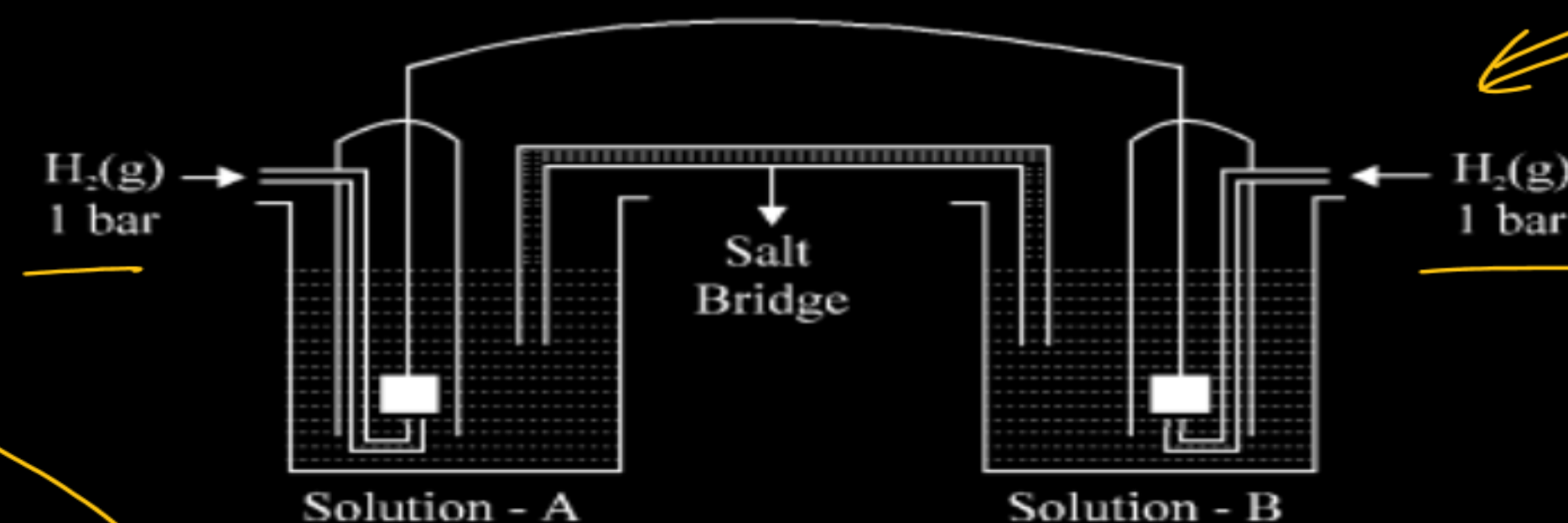
$\Rightarrow a^3 = \frac{63.5}{5.4 \times \sqrt{2}} \Rightarrow a = \left(\frac{63.5}{5.4 \times \sqrt{2}} \right)^{1/3}$

(A) $\left(\frac{635}{5.4 \times \sqrt{2}} \right)^{1/3} \text{ \AA}$ (B) $\left(\frac{635 \times \sqrt{2}}{5.4} \right)^{1/3} \text{ \AA}$ (C) $\left(\frac{635}{16.2 \times \sqrt{2}} \right)^{1/3} \text{ \AA}$ (D) $\left(\frac{635 \times \sqrt{2}}{16.2} \right)^{1/3} \text{ \AA}$

unit cell.



Using data provided, find the EMF of shown Galvanic cell (in volt):



$$pOH = pK_b + \log \frac{s}{b}$$

$$pOH = 5 + \log \frac{0.1}{0.1}$$

$$[OH^-] = 10^{-5} M$$

$$[H^+] = c_1 = 10^{-9} M$$

Solution A is 0.1 M each in NH_4OH and NH_4Cl and solution B is 0.1 M CH_3COONa .

[Given : $K_a(CH_3COOH) = 10^{-5}$, $K(NH_4OH) = 10^{-5}$ and $\frac{2.303RT}{F} = 0.06$ volt]

$$E = 0 - \frac{0.06}{1} \log \frac{c_1}{c_2} \quad \text{--- (1)}$$

eq. (1),

$$E_{cell} = 0 - \frac{0.06}{1} \log \frac{10^{-9}}{10^{-9}}$$

$$= 0 \quad \text{Ans}$$

$[H^+] = c_2 = 10^{-9} M$

Salt hydrolysis.

$$pH = \frac{1}{2} (pK_w + pK_a + \log c)$$

$$= \frac{1}{2} (14 + 5 + \log 0.1)$$

$$= \frac{1}{2} (19 - 1) = 9$$

A bottle of oleum is labelled as 109%. Which of the following statement is/are correct for this oleum sample?



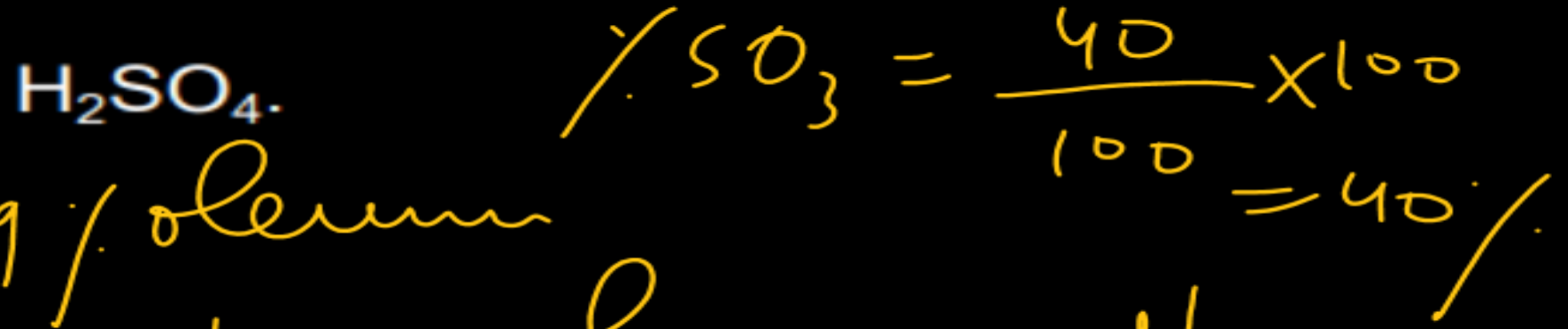
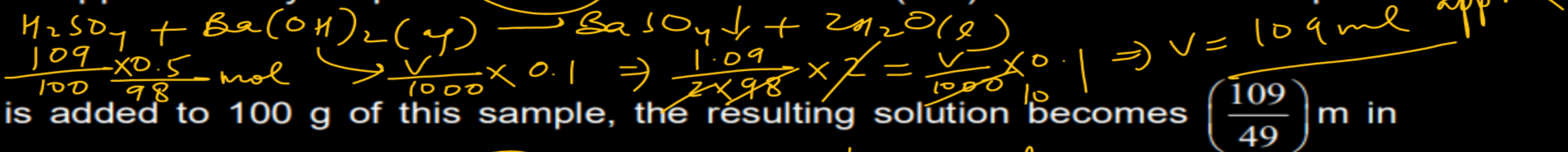
SO_3 dissolved in H_2SO_4 oleum

(A) It contains 40% of free SO_3 by weight.

(B) 1.0 g of this sample approximately requires 22.25 ml of 0.5 M-NaOH solution for complete neutralization.

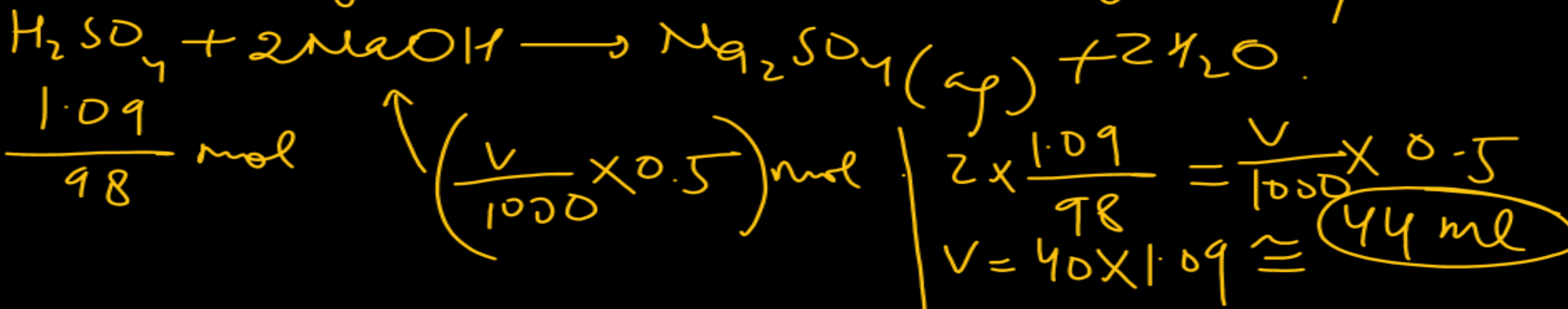
(C) 0.5 g of this sample approximately requires 11.12 ml of 0.1 N-Ba(OH)₂ solution for complete neutralization.

(D) When 500 g water is added to 100 g of this sample, the resulting solution becomes $\left(\frac{109}{49}\right)$ m in

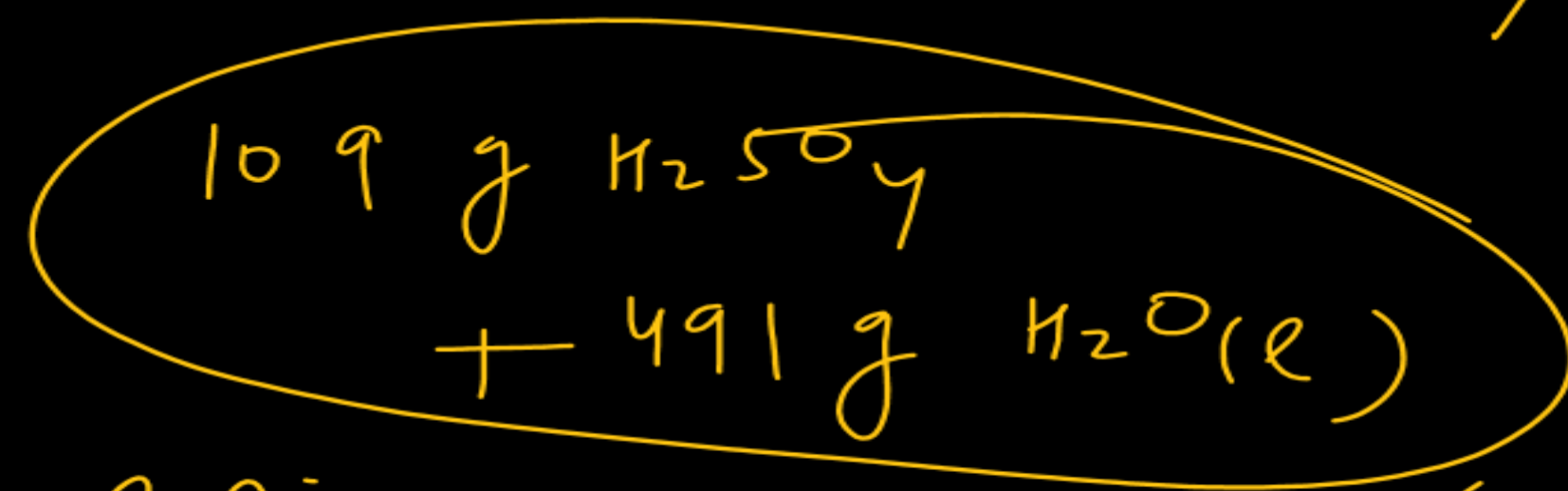


(A) 109% oleum
 100 g oleum sample
 $\swarrow$
 40 g SO_3 (Free SO_3) $\searrow$ 60 g H_2SO_4

(B) 100 g sample + water (enough) $\rightarrow$ 109 g max. H_2SO_4
 $\therefore 1 \text{ g} = 1.09 \text{ g } \text{H}_2\text{SO}_4$



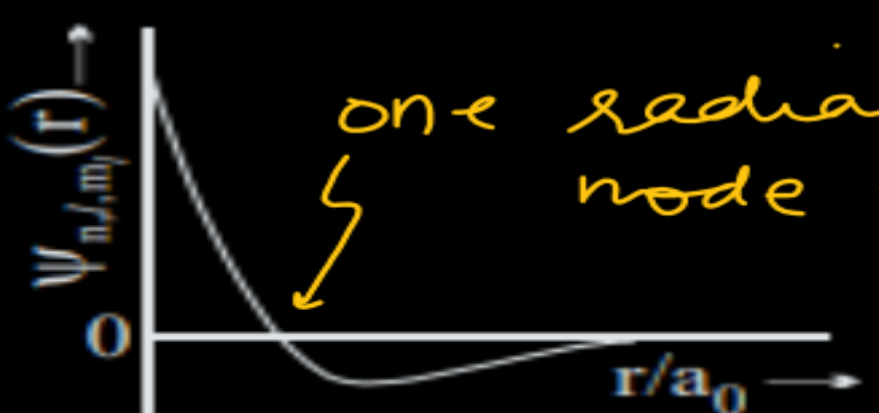
① 300 g water in 100 g oleum (109%)



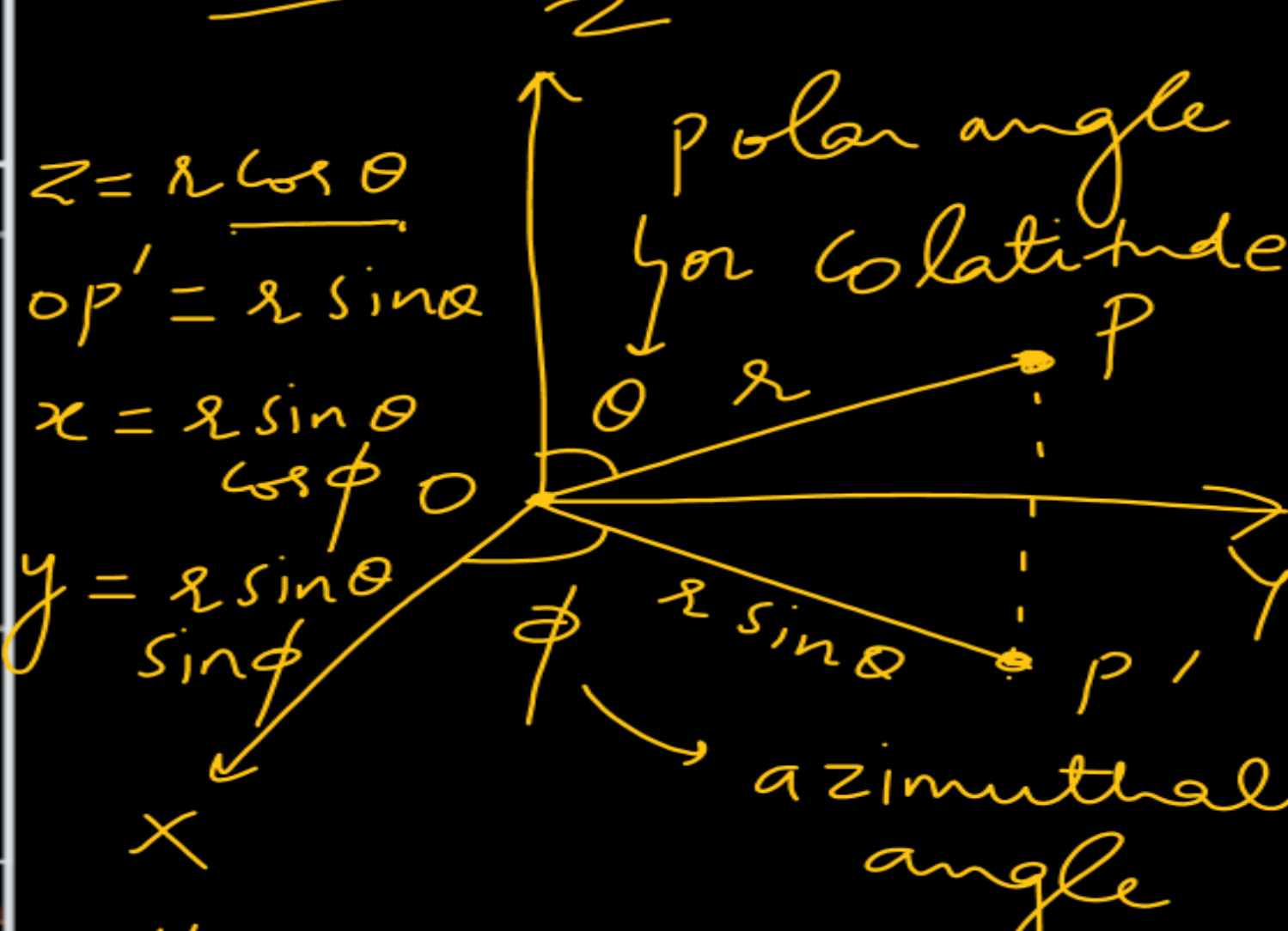
$$\text{molality of H}_2\text{SO}_4 = \frac{\left(\frac{109}{98}\right) \text{ mol}}{0.491 \text{ kg}}$$

$$= \frac{109}{98 \times 0.491} = \frac{109}{49} = 2.22 \text{ molal}$$

Answer Q.6, Q.7 and Q.8 by appropriately matching the information given in the three columns of the following table.

The wave function, ψ_{n,l,m_l} is a mathematical function whose value depends upon <u>spherical polar coordinates</u> (r, θ, ϕ) of the electron and characterized by the quantum numbers n, l and m_l . Here r is distance from nucleus, θ is colatitude and ϕ is azimuth. In the mathematical functions given in the Table, Z is atomic number and a_0 is Bohr radius.		
Column 1	Column 2	Column 3
(I) 1s orbital $(i), (Q), (R), (S)$	(i) $\psi_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{3}{2}} e^{-\left(\frac{Zr}{a_0}\right)}$	(P) 
(II) 2s orbital $(ii), (P), (Q), (R), (S)$	(ii) One radial node	(Q) Probability density at nucleus $\propto \frac{1}{a_0^3}$ $r \rightarrow 0$
(III) <u>2p_z</u> orbital $2p_z, (iii), (iv)$	(iii) $\psi_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{5}{2}} r e^{-\left(\frac{Zr}{2a_0}\right)} \cos \theta$ $\rightarrow p_z$	(R) Probability density is maximum at nucleus
(IV) 3d _{xy} orbital X	(iv) xy-plane is a nodal plane	(S) Energy needed to excite electron from $n=2$ state to $n=4$ state is $\frac{27}{32}$ times the energy needed to excite electron from $n=2$ state to $n=6$ state

JEE Adv - 2017



$z = r \cos \theta$
 $OP' = r \sin \theta$
 $x = r \sin \theta \cos \phi$
 $y = r \sin \theta \sin \phi$

polar angle θ
 for colatitude
 azimuthal angle ϕ

$\psi(n, l, m) = Y_l^m \times Y(\theta, \phi)$
 s orbital $\rightarrow$ 4 does not depend on θ, ϕ ,
 correct for all single e^- species.

General equation of ψ_r

$$\psi_r = \kappa \underbrace{r^l}_{\text{(a polynomial of } r \text{ with degree } n-l-1)} \times e^{-r/\kappa'}$$

$$\psi_{r1s} = \underbrace{\kappa}_{(l=0)} e^{-r/\kappa'} \quad \left(\kappa \propto \left(\frac{r}{a_0} \right)^{3/2} \right)$$

$$\psi_{r2s} = \underline{\kappa} (\kappa' - r) e^{-r/\kappa''} \quad (l=0)$$

$$\psi_r(zp) = \kappa \underbrace{r^1}_{(l=1)} e^{-r/\kappa''}$$

$$(l=1) \quad \psi_r(3d) = \kappa \underbrace{r^2}_{(l=2)} e^{-r/\kappa'}$$

For single e^- species:

$$\frac{E_4 - E_2}{E_6 - E_2} = \frac{13.6 Z^2 \left(\frac{1}{2^2} - \frac{1}{4^2} \right)}{13.6 Z^2 \left(\frac{1}{2^2} - \frac{1}{6^2} \right)} = \frac{\frac{3}{16} \times \frac{36}{8}}{\frac{16}{8}} = \frac{27}{32}$$

$$\psi(\theta, \phi) \text{ for } zp \rightarrow f(\cos \theta)$$

$$\psi(\theta, \phi) \text{ for } zp \rightarrow f(\sin \theta \cos \phi)$$

$$\psi(\theta, \phi) \text{ for } zp \rightarrow f(\sin \theta \sin \phi)$$

$$\psi_{n,l,m} = \psi_r \times \psi(\theta, \phi)$$

⇒ For He^+ ion, the only INCORRECT combination is

(A) (I) (i) (R)

(B) (II) (ii) (Q)

(C) (I) (i) (S)

☒ (D) (I) (iii) (R)

⇒ For the given orbital in column 1, the only CORRECT combination for any hydrogen-like species is

(A) (I) (ii) (S)

(B) (IV) (iv) (R)

(C) (III) (iii) (P)

☒ (D) (II) (ii) (P)

X

X

X

⇒ For hydrogen atom, the only CORRECT combination is

(A) (II) (i) (Q)

(B) (I) (iv) (R)

(C) (I) (i) (P)

☒ (D) (I) (i) (S)

X

X

X